

LOCALIZATION AND QUANTIFICATION OF PMAP23 DURING *E. COLI* KILLING

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ABSTRACT

Antimicrobial peptides (AMPs) rapidly kill bacteria by permeabilizing their membranes, yet their spatial distribution throughout the killing process remain largely unresolved. In our previous kinetic study, dansyl-labelled PMAP23 revealed four sequential events during its interaction with *E. coli*: outer-membrane binding, outer-membrane permeabilization, onset of bacterial killing, and inner-membrane permeabilization. Here, we extend this kinetic framework by investigating the localization and mobility of ATTO647-labelled variant of PMAP23 before and after bacterial killing by using two-photon NAD(P)H phasor-FLIM, confocal microscopy, stimulated emission depletion microscopy (STED), and raster image correlation spectroscopy (RICS). NAD(P)H-FLIM revealed concentration-dependent changes in the mean bound NAD(P)H fraction, consistent with bacterial killing. Mid-plane STED imaging resolved two distinct peptide distributions: a predominant membrane-associated population and a smaller peptide fraction showing intracellular accumulation, both occurring in membrane-permeabilized bacteria as shown by Sytox Green co-staining. Interestingly, RICS revealed localization-dependent peptide mobility regimes across both permeabilizing and non-permeabilizing peptide concentrations. Bacteria displaying predominantly membrane-associated PMAP23 showed both a slow component ($D \sim 0.2\text{--}0.3 \mu\text{m}^2 \text{s}^{-1}$), and a much faster mobile population, compatible with freely diffusing peptide molecules in the cytosol. In contrast, bacteria showing extensive intracellular accumulation displayed only a slow component, suggesting that intracellularly accumulated peptide is not freely diffusing but is instead strongly confined. Remarkably, rare bacteria displaying intracellular peptide accumulation together with slow peptide mobility were also detected at non-permeabilizing concentrations, including in the nanomolar range. Overall, our findings indicate that PMAP23 can follow different intracellular paths within the bacterial population. A fraction of the peptide appears to enter the cytosol and remain highly mobile, whereas strong intracellular accumulation is associated with markedly reduced mobility, consistent with sequestration by cellular components. This raises the possibility that severely damaged or dead bacteria act as peptide sinks, decreasing the amount of AMP available to surrounding cells and potentially promoting population-level tolerance.